

Terminal Video Kit

Terminal ki asli commands + Hindi narration + line highlight = teaching video. Team ke liye poori guide.

Repo

<https://github.com/mohitsoniv/terminal-video-kit>

Shuru karne ke liye teen commands (details section 2 aur 3 mein):

```
git clone https://github.com/mohitsoniv/terminal-video-kit.git
cd terminal-video-kit
./build.sh topics/http          # setup ho chuka ho to pehli video ban jayegi
```

1. Yeh kit karta kya hai

Aap ek command likhte ho aur kya bolna hai woh likhte ho. Kit baaki sab khud karta hai:

1. **VHS** aapki command ko ek asli terminal mein chalata hai aur screen record karta hai. Output nakli nahi hota, jo sach mein chala wahi dikhta hai.
2. **edge-tts** har sentence ki Hindi awaaz banata hai.
3. **highlight.py** hisaab lagata hai ki kaunsa sentence bolte waqt screen ki kaunsi line highlight honi chahiye.
4. **ffmpeg** sab jodkar ek MP4 banata hai, peelee highlight box aur popup ke saath.

Nateeja: ek video jisme narrator jis line ki baat kar raha hai, wahi line us waqt highlight hoti hai. Aur zaroori jaankari (jaise flag ka matlab) neeche popup box mein aati hai.

2. Kit lena (GitHub se)

```
git clone https://github.com/mohitsoniv/terminal-video-kit.git
cd terminal-video-kit
ls          # build.sh highlight.py  calib/  new-topic.sh  README.md  topics/
```

Repo mein sirf source files hain: scripts, commands aur narration. Videos, frames, mp3 aur `calib/calib.json` jaan-boojh kar nahi rakhe gaye, kyunki woh har machine ke apne hote hain aur repo bhaari kar dete hain. Calibration pehli build pe khud ho jaata hai.

Kit mein pehle se kuch topics maujood hain, jinhe dekhkar apna topic likhna aasaan ho jata hai: `topics/http`, `topics/dbms-00-intro`, `topics/dbms-01-basics`, `topics/dbms-02-crud`.

Naya kaam shuru karne se pehle hamesha:

```
git pull
```

3. Ek baar ka setup (har nayi machine pe)

```
sudo apt update
sudo apt install ffmpeg tttyd netcat-openbsd pipx
pipx install edge-tts
pipx ensurepath

# VHS (Charm ke apt repo se)
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://repo.charm.sh/apt/gpg.key | sudo gpg --dearmor -o /etc/apt/keyrings/charm.gpg
echo "deb [signed-by=/etc/apt/keyrings/charm.gpg] https://repo.charm.sh/apt/ * *" | sudo tee /etc/apt/sources.list.d/charm.list
sudo apt update && sudo apt install vhs
```

Naya terminal kholo (PATH ke liye) aur check karo:

```
vhs --version && edge-tts --version && ffmpeg -version | head -1
```

tttyd ki service band kar dein

Install ke baad tttyd ek background service chalu kar deta hai jo zaroori nahi hai: `sudo systemctl disable --now tttyd`

RAM kam ho to swap bana lein

VHS ke andar Chrome chalta hai. 2 GB se kam free RAM pe recording chup-chaap ruk sakti hai. `sudo fallocate -l 2G /swapfile && sudo chmod 600 /swapfile && sudo mkswap /swapfile && sudo swapon /swapfile`

Calibration (apne aap, sirf pehli baar)

Pehli build pe kit terminal pe ek test pattern chhap kar naapta hai ki ek line kitni oonchi hai aur screen mein kitne columns aate hain. Yeh `calib/calib.json` mein save ho jaata hai aur dobara nahi hota. Isi se highlight box sahi line pe baithta hai. **Yeh file har machine ki apni hoti hai, isliye kisi aur ko copy karke mat dena.**

4. Kit ka dhaancha: kya chhoona hai, kya nahi

Path	Kya hai	Aapko?
<code>build.sh</code>	Poora kaam kram se chalata hai	Haath mat lagao
<code>highlight.py</code>	Highlight, awaaz aur timing ka dimaag	Haath mat lagao
<code>calib/</code>	Calibration ka data	Haath mat lagao
<code>new-topic.sh</code>	Naya topic folder banata hai	Chalana hai
<code>topics/<naam>/steps/</code>	Commands (ek step = ek file)	Aap likhoge
<code>topics/<naam>/narration/</code>	Kya bolna hai + highlight patterns	Aap likhoge
<code>topics/<naam>/setup.sh</code>	Video se pehle ki taiyari (optional)	Zarurat pe
<code>topics/<naam>/cleanup.sh</code>	Video ke baad safai (optional)	Zarurat pe
<code>topics/<naam>/build/</code>	Kit ka apna kaam (frames, mp3, clips)	Chhedna nahi, delete kar sakte ho
<code>topics/<naam>/<naam>.mp4</code>	Final video	Yahi deliverable hai

Ek line mein
Har naye topic ke liye sirf do folder bharne hain: `steps/` aur `narration/` .

5. Nayi video banane ka poora tareeka

Kadam 1 — topic banao

```
cd ~/terminal-video-kit
./new-topic.sh docker-basics
```

Kadam 2 — commands likho (steps/)

Har step ki apni file. Naam hamesha number se: 01 , 02 , 03 ... isi kram mein video banti hai.

(a) Ek command wala step — steps/01.cmd

```
# Output aane mein 3 second se zyada lage to yeh line upar likho:
# wait 6
docker ps
```

Is file mein sirf **ek** command honi chahiye, ek hi line mein. `&&` se do commands jodna theek nahi, unhe alag steps banao.

(b) Interactive session wala step — steps/02.session

```
# wait 5
sudo -u postgres psql
\l
\c shopdb
\dt
\q
```

Pehli line woh program jo khulega, baaki lines uske andar chalengi. Video mein har line alag-alag type hoti dikhegi, asli prompt ke saath. Ek step mein `.cmd` ya `.session` , dono nahi.

(c) Reset script — steps/02.pre (optional)

```
psql -d shopdb -q -c "DROP TABLE IF EXISTS demo;"
psql -d shopdb -q -c "CREATE TABLE demo (id INT);"
```

Yeh step se pehle chupchap chalti hai, video mein nahi dikhti. Iski zarurat kyun padti hai, woh section 8 mein.

Kadam 3 — narration likho (narration/NN.txt)

Har step ki narration file ka naam step jaisa hi: 01.cmd ke liye narration/01.txt . Har line ka format:

```
pattern ||| kya bolna hai ||| popup text (optional)
```

Asli example (yahan Hindi ko Roman mein dikhaya gaya hai; file mein sentence Devanagari mein likhein):

```
||| [Ab hum database se jud rahe hain.]
^$ sudo ||| [Yeh command postgres user banke psql kholti hai.] ||| sudo -u postgres psql = postgres user banke connect
postgres=# ||| [Yeh psql ka prompt hai.]
^ Database ||| [Hum postgres naam ke default database mein hain.] ||| \conninfo = connection ki jaankari
```

Pehli line ka pattern khaali hai (line ke shuru mein ek space), isliye us sentence pe koi highlight nahi aayega.

Hissa	Kaam
pattern	Screen ki jis line se match karega, woh line us sentence ke dauraan highlight hogi. Khaali chhodo to koi highlight nahi (shuruaati ya aakhri baat ke liye theek).
sentence	Jo bola jayega. Devanagari mein likho, technical shabd (psql, foreign key) English mein.
popup	Screen mein neeche peelee box mein dikhega. Sirf English/Roman mein likho — Devanagari yahan theek nahi dikhta.

Kadam 4 — patterns check karo (video banne se pehle)

```
cd ~/terminal-video-kit
bash topics/docker-basics/setup.sh          # agar setup.sh hai
python3 highlight.py check topics/docker-basics
```

Yeh har command chala kar dikhata hai ki kaunsa pattern kis line se mila. `MISS` ka matlab hai us sentence pe highlight nahi aayega. Sab `OK` hone tak build mat chalao — 15 minute ka build karke galti pata chalna mehnga padta hai.

Kadam 5 — video banao

```
./build.sh topics/docker-basics
```

Video milegi: topics/docker-basics/docker-basics.mp4

6. Commands ki cheat sheet

Command	Kaam
<code>./new-topic.sh <naam></code>	Naya topic folder
<code>python3 highlight.py check topics/<naam></code>	Sirf patterns check (video nahi banti)
<code>./build.sh topics/<naam></code>	Video banao (bane hue steps skip honge)
<code>FORCE=1 ./build.sh topics/<naam></code>	Saare steps dobara banao
<code>VOICE=hi-IN-SwaraNeural ./build.sh ...</code>	Female awaaz (default: hi-IN-MadhurNeural)
<code>FPS=30 ./build.sh ...</code>	Zyada smooth recording (dheemi build, default 15)
<code>rm -rf topics/<naam>/build/*-frames</code>	Jagah khali karo (clips bache rahenge)

Ek step badalna ho to sirf uski narration ya command badlo aur build dobara chalao — baaki steps skip ho jayenge, kaam 2 minute mein.

7. Video ka rang-roop

- Prompt aapke apne machine ka dikhta hai, rangon ke saath: `keen@keen-pc:~$`
- Highlight: peela box, ek waqt mein sirf usi line pe jiski baat ho rahi hai. Agla sentence shuru hote hi pichhla box hat jaata hai.
- Popup: screen mein neeche peela box, sirf us sentence ke dauraan.
- Command khatam hone ke baad screen ruki rehti hai jab tak narration khatam na ho, isliye viewer ko padhne ka waqt milta hai.
- Screen: 1280x720, lagbhag 91 columns aur 27 lines.

8. Niyam, jinke bina video bigdegi

Har command build ke dauraan 2 baar chalti hai

Ek baar VHS ki recording ke liye, ek baar highlight ka hisaab lagane ke liye. Isliye:

- Data badalne wali commands (`INSERT` , `UPDATE` , `DELETE`) mein `.pre` reset script zaroori hai, warna doosri baar output alag aayega.
- System badalne wali commands (`rm` , `apt install` , `systemctl restart` , `kubectl delete`) bilkul mat daalo. Build inhe khud rok deta hai. Pakka safe ho to `ALLOW_DANGEROUS=1 ./build.sh ...`

Niyam	Wajah
Kabhi khatam na hone wali commands nahi (<code>top</code> , <code>vim</code> , <code>less</code> , <code>tail -f</code> , <code>watch</code>)	Recording atak jayegi. Kit 30 second baad rok deta hai.
<code>sudo</code> password maangega to build ruk jayega	Recording ke andar koi type karne wala nahi hota. Hal section 8 mein.
Output 25 line se lamba na ho	Upar ki lines scroll hokar gayab ho jayengi. <code> head -n 20</code> ya <code> grep</code> lagao.
Secrets kabhi screen pe na aayein	Password, token, IP, client ka naam — jo screen pe hai woh video mein hai. Demo data pe kaam karo.
Steps 6-8 se zyada na rakho	10 steps ka build 15-20 minute leta hai, aur video bhi lambi ho jaati hai.

9. Aam dikkatein aur unka hal

Kya dikha	Matlab aur hal
MISS <code><pattern></code>	Woh line output mein hai hi nahi. Asli output dekho aur pattern theek karo. Tool ka version badalne se wording badal jaati hai (jaise <code>psql 18</code> mein "Did not find any tables").
NOTE: N rows ka output, screen 27 rows ki	Upar ki lines scroll ho jayengi. Output chhota karo ya step ko do steps mein todo.
<code>sudo: a password is required</code>	Recording ke waqt ke liye seemit rule lagao (section 9.1), ya <code>sudo</code> wali command hatao.
Port 8080 pe pehle se kuch chal raha hai	Doosra port chuno, chalti hui service band mat karo.
Build atka hua lagta hai	<code>ls topics/<naam>/build/*-frames/ wc -l</code> chala ke dekho. Number badh raha hai to chal raha hai.
Video player mein nahi chalti	Pehle <code>ffplay <file></code> se dekho. Chal jaye to player ke codecs ki kami hai: <code>sudo apt install vlc</code>
Invalid command: home (VHS)	VHS absolute path nahi samajhta. Kit khud relative path deta hai; tape haath se mat likho.
Highlight box ek line upar/neeche	Calibration dobara karwao: <code>rm calib/calib.json</code> phir build chalao.

9.1 sudo wali commands recording mein

Sirf tab, jab video mein `sudo` dikhana hi zaroori ho. Rule sirf ek command tak seemit rakho, aur recording ke turant baad hata do.

```
# lagao
echo "$(id -un) ALL=(postgres) NOPASSWD: /usr/bin/psql" | sudo tee /etc/sudoers.d/vhs-recording
sudo chmod 0440 /etc/sudoers.d/vhs-recording
sudo visudo -c -f /etc/sudoers.d/vhs-recording

# build ke baad hata do
sudo rm /etc/sudoers.d/vhs-recording
```

Jab tak yeh rule laga hai, aapke user ko bina password us command ki पूरी ताकत मिल जाती है. Shared machine पे सोच-समझ कर लैगो.

10. Achhi video ke liye sujhav

- **Ek video, ek vishay.** 6-8 steps, 4-6 minute. Lambi list ko do videos mein todo.
- **Shuruaat mein batao ki is video mein kya karenge**, aur pichhli video ka ek line ka reference do.
- **Galti jaan-boojh kar dikhao.** Foreign key ka error, ya bina `WHERE` wala `UPDATE` — error wale steps sabse zyada yaad rehte hain.
- **Output ka har hissa alag sentence mein samjhao**, taaki highlight ek-ek line pe chale, poore block pe nahi.
- **Popup sirf flags aur short definitions ke liye**, poora vakya nahi.
- **Data ek series mein ek jaisa rakho.** Pichhli video mein Mohit ka order 500 tha to agli video mein bhi 500 hi rahe.

11. Seemayein, jo pehle se jaan lena behtar hai

- **edge-tts online hai** aur Microsoft ki service par nirbhar hai. Woh band ho to awaaz ka hissa ruk jayega; tab Piper jaisa offline TTS lagana padega.
- **Apni awaaz chahiye** to voice cloning ke models (XTTS, F5-TTS) aam taur pe non-commercial license pe aate hain, aur bina GPU ke bahut dheeme hain. Commercial kaam ke liye ElevenLabs jaisa paid service, ya khud har sentence record karna.
- **Linux/macOS ke liye hai.** Windows wale WSL mein chalayein.
- **Jo commands recording mein nahi chal sakti** (install, service restart, poora cluster setup) unke liye alag throwaway VM ya container chahiye.

12. Team ke saath kaam karna (Git)

Kit yahan rehta hai: <https://github.com/mohitsoniv/terminal-video-kit>

Apna topic add karna

```
cd terminal-video-kit
git pull
./new-topic.sh redis-basics
# steps/ aur narration/ likho, check chalao, video banao
git add topics/redis-basics
git commit -m "redis basics video ka topic"
git push
```

Video khud commit mat karo — sirf source (commands aur narration). MP4 Drive ya kisi video platform pe rakho.

Kya commit nahi hota

`.gitignore` pehle se yeh sab rokta hai: `topics/*/build/`, saari `*.mp4` aur `*.mp3`, `calib/calib.json` aur `__pycache__/`. Push se pehle `git status --short` zaroor dekh lo — sirf text files dikhni chahiye.

Push se pehle ek jaanch

```
grep -rniE "password|secret|token" topics/ --include="*.cmd" --include="*.session" --include="*.sh"
```

Commands aur output video mein aur repo mein, dono jagah jaate hain. Client ka naam, internal hostname, IP ya token kahin bhi nahi hona chahiye. Repo private hai to bhi yahi niyam rakho.

build.sh ya highlight.py badalna ho to

Yeh dono poori team ke liye common hain. Seedha `main` pe badalne ke bajaye alag branch banao aur review karwao, warna kisi ek ki video ka fix doosre ki video tod sakta hai.

Ek nazar mein poora kaam

`./new-topic.sh <naam>` → `steps/*.cmd` likho → narration likho → `highlight.py` check → `./build.sh topics/<naam>` → MP4 taiyar.